

Multiple Choice (10 marks)

1. Statement 1 | For $n > 1$, the set $\{1, 2, \dots, n-1\}$ is a group under multiplication modulo n . Statement 2 | There is an integer x such that $63x \bmod 100 = 1$.
 - (a) False, True
 - (b) False, False and x is less than 100
 - (c) False, True and x is an odd number
 - (d) True, True and x is greater than 100
 - (e) False, True and x is a prime number
2. Both A, B are n -by- n matrices with $\text{rank}(A)=n$, $\text{rank}(A*B)=0$. What is $\text{rank}(B)$?
 - (a) 0.0
 - (b) Undefined
 - (c) 2
 - (d) $n-1$
 - (e) $n+1$
3. A relative maximum value of the function $y = (\ln x)/x$ is
 - (a) $e/2$
 - (b) 1
 - (c) e^2
 - (d) $2/e$
 - (e) $1/2$
4. Is there exist a holomorphic function f on the unit disk $B(0, 1)$ (boundary excluded) such that $f(B(0, 1)) = C$? Here C is the complex space. Return 1 for yes and 0 for no.
 - (a) 2.5
 - (b) -2.0
 - (c) 3.0
 - (d) 2.0
 - (e) 1.5
5. What is $(6^8 3 + 8^8 3) \bmod 49$?
 - (a) 42
 - (b) 14

- (c) 35
- (d) 21
- (e) 49

True / False (10 marks)

Answer True or False

6. True or False: The sequence 17,400, 116,633, 161,455, 167,213 is correctly ordered from least to greatest.
7. True or False: The number of positive integers that are factors of both 120 and 40 is 4.
8. True or False: The improper fraction representation of $4\frac{4}{5}$ is $\frac{26}{5}$.
9. True or False: The total cost of a 6-lb bag of cat food at \$.47/lb is \$6.47.
10. True or False: The numbers 1,449 and 1,549 both round to 1,500 when rounded to the nearest hundred.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the domain of the function $f(x) = \frac{x+6}{\sqrt{x^2-3x-4}}$. Analyze the constraints imposed by the denominator and determine the valid intervals for x .
12. Given that it takes 2 J of work to stretch a spring from its natural length of 30 cm to 42 cm, determine how far a 30 N force will stretch the spring beyond its natural length. Provide a detailed analysis of your calculations and reasoning.

End of Paper